**Godfred Opoku Ababio**
*Research Portfolio*

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## EDUCATION

Ph.D., Structural & Earthquake Engineering, University of California, Los Angeles, CA (2027)

M.S., Structural & Earthquake Engineering, University of California, Los Angeles, CA (2024)

B.S., Civil Engineering, Kwame Nkrumah University of Science and Technology (KNUST) (2021)

## RESEARCH

### ***Research Interests***

- Data-Driven Performance-Based Engineering
- Seismic Resilience and Risk Mitigation Strategies
- Seismic Hazard and Risk Assessment
- Building Post-Earthquake Functional Recovery Modeling
- Earthquake Risk Sharing Through Community-Based Risk Pooling

### ***Projects***

**2023-Present** Lead Graduate Student Researcher, Quantifying and Mitigating Seismic Risk Disparities in Residential Communities.

My research advances earthquake resilience through the integration of seismic risk assessment and evaluating different structural mitigation and community-based risk-sharing strategies. In this work, I investigate spatial and socioeconomic disparities in seismic retrofit adoption and the associated risk reduction benefits. I am also developing probabilistic frameworks to quantify the effectiveness of seismic retrofitting in reducing economic losses and improving post-earthquake functional recovery. A key focus of my work is understanding how the choice of performance metrics and analytical methods influences conclusions about disparities in the distribution of seismic risk-reduction benefits. Additionally, I explore stochastic models of reciprocal insurance exchanges as alternative mechanisms for community-based earthquake risk sharing.

**2022–2023** Research Assistant, Development of Machine Learning Models for Failure Mode Classification of Reinforced Concrete Rectangular Columns.

Developed and optimized machine learning models, including SVM, ANN, and Decision Trees, to classify shear, flexure-shear, and flexural failure modes.

## EXPERIENCE

**2023-Present** Graduate Student Researcher and Teaching Assistant,
University of California, Los Angeles

**2022-2023** Research and Teaching Assistant, KNUST

## HONORS AND AWARDS

- Graduate fellowship by the UCLA Ralph J Bunche Center, 2026-2027
- Received Vilas and Ingrid Mujumdar Registration Grant for the 13NCEE, Portland, Oregon, 2026
- Selected Participant of the Structural Engineers Association of California (SEAOC) Pathways Cohort, 2025
- Selected Participant of the NHERI Strategic Program to Accelerate Researchers in Computing (SPARC) and awarded a travel grant to the program at UT Austin, 2025
- Summer Mentored Research Fellowship. University of California, Los Angeles, 2024.
- Graduate fellowship by the UCLA Ralph J Bunche Center, 2023 - 2025

## PUBLICATIONS

Note: names with "*" are current advisor, and with "+" are past advisors.

1. Ababio, G., Burton, H., Derakhshan, S., Galvis, F., & Issa, O. (2026). Thirty years of earthquake retrofit in the city of Los Angeles: Trends, drivers, and implications for seismic resilience. International Journal of Disaster Risk Reduction, 106177.

### ***In-preparation***

2. **Ababio, G**., Burton, H*., (2026+). "Metrics to Quantify Risk Reduction Benefits Derived from Cripple Wall Retrofits in the City of Los Angeles" (in preparation).

3. **Ababio, G**., Burton, H*., (2026+). "The Role of Loss Dependence in the Performance of Reciprocal Insurance Exchanges" (in preparation).

## PRESENTATIONS

"Earthquake Risk Sharing through Community-Based Reciprocal Exchanges," 13NCEE, Portland, Oregon, July 2026.

"Earthquake Risk Sharing through Community-Based Reciprocal Exchanges," NHERI *SimCenter Symposium*, Berkeley, CA, May 2026.

"Thirty Year of Earthquake Retrofit Activity in the City of Los Angeles: Trends, Drivers, and Implications for Seismic Resilience" *SEAOC Convention*, San Diego, CA, September 2025.

"Sociodemographic Disparities in the Risk Reduction Benefits Derived from Cripple Wall Retrofits in the City of Los Angeles," NHERI *SimCenter Symposium*, Los Angeles, CA, February 2025.

## TEACHING

### ***Co-Instructor***

*CE 151 – Elementary Structures (January 2022-April 2022 – KNUST, Ghana)*

Types of structures and loads, Analysis of statically determinate structures and trusses, internal loadings developed in structural members (Axial force, Shear force and Bending Moment Diagrams), Cables and Arches, Influence lines

***Teaching Assistant***

*C&EE 144 – Structural System Design (Spring 2026, UCLA)*

Seismic Design Criteria, Wind Design Criteria, Reinforced Concrete Design, Mass Timber, Wood Design

*CE 255 – Theory of Structures (June 2022 – August 2022– KNUST, Ghana)*

Slope and displacement of structures (Conjugate beam method, Moment-area method, virtual work methods for beams and trusses)

*CE 257 – Python Programming for Engineers (June 2022 – August 2022– KNUST, Ghana)*

Control Structures and functions, Working with NumPy for Numerical Computing, Data Handling with Pandas, Data Visualization with Matplotlib

*CE 351 – Structural Analysis (January 2022-April 2022 – KNUST, Ghana)*

Displacement Method of Analysis (Moment distribution, Slope-displacement Equation), Three-Moment Equation, Castigliano's Second Theorem.

## ADVISING

- Co-supervised an undergraduate student (Gantt Essence) from Howard University as part of UCLA-HBCU Summer program to evaluate the effect of retrofit on post-earthquake functional recovery of single-family dwellings in LA city, 2026

- Co-supervised an undergraduate student (Ammaarah Gage) as part of my graduate fellowship by the Ralph J Bunche Center for African American studies to develop equitable earthquake risk and resilience solutions, 2025

## VOLUNTEERING

- Participated in a community fire safety initiative by replacing residential smoke detectors to enhance fire preparedness and promote community safety. American Red Cross, Pacific Palisades, CA, July 2026
- Assisted in sorting, organizing, and packaging food donations for distribution to individuals and families experiencing food insecurity. Los Angeles Regional Food Bank, Los Angeles, CA, August 2026
- Participated in community outreach initiatives by preparing and distributing meals to individuals experiencing homelessness and food insecurity. Skid Row, Los Angeles, CA